FinGenIQ ENTERPRISE AUDIT

Comprehensive Development Cost, Full Infrastructure Stack & High Concurrency (1k – 10k CCU) Audit

Generated: August 2026
Stack: Next.js 16 / React 19 / Three.js 3D
Target Scale: 1,000 to 10,000 Concurrent Users (CCU)

INITIAL DEV COST
$22,000 – $48,000
Complete MVP to Prod Build

1,000 CCU OPEX
$280 – $780 /mo
~50k–100k Registered Users

10,000 CCU OPEX
$1,800 – $5,900 /mo
~500k–1,000,000 Registered

UNIT COST / USER
$0.006 – $0.018
Per active user / month

1. Full Development Cost Breakdown (CapEx)

Engineering effort required to develop, design, test, and deploy the entire FinGenIQ platform including Three.js WebGL visualizations, interactive learning player, assessment engine, cryptographic certificate verification, admin provisioning, and multi-tier authentication.

WORKSTREAM / COMPONENT	SCOPE OF WORK	HOURS	RATE ($/HR)	TOTAL COST
UI/UX Design & 3D Assets	Design system, dark/light tokens, Three.js 3D models, shader animations, responsive layouts	70 – 100 hrs	$50 – $80	$3,500 – $8,000
Frontend & WebGL Canvas	Next.js 16 App Router, React 19 client components, Three.js interactive canvas, state management	120 – 160 hrs	$65 – $100	$7,800 – $16,000
Backend, DB & Auth Engine	Admin-provisioned auth, session management, bcrypt/argon2, audit logs, renewal system, rate limiting	90 – 130 hrs	$70 – $110	$6,300 – $14,300
Courseware & Assessments	Interactive step-by-step player, capstone projects, grading, SHA-256 certificate hashing engine	60 – 80 hrs	$60 – $90	$3,600 – $7,200
DevOps, CI/CD & Security	Containerization (Docker), PgBouncer, Redis clustering, Cloudflare WAF, load testing & Pen-testing	40 – 60 hrs	$75 – $120	$3,000 – $7,200
TOTAL ESTIMATED BUILD EFFORT		**380 – 530 hrs**	**—**	**$24,200 – $52,700**

2. Understanding 1,000 to 10,000 Concurrent Users (CCU)

Concurrent Users (CCU) represents users interacting with the platform simultaneously at the same second. This determines CPU, memory, socket connections, and database throughput requirements.

METRIC	1,000 CONCURRENT USERS (CCU)	5,000 CONCURRENT USERS (CCU)	10,000 CONCURRENT USERS (CCU)
Equivalent Registered Base	50,000 – 100,000 users	250,000 – 500,000 users	500,000 – 1,000,000 users
Peak Requests per Second (RPS)	500 – 1,500 RPS	2,500 – 7,500 RPS	5,000 – 15,000 RPS
Database Queries per Sec (QPS)	1,500 – 4,500 QPS	7,500 – 22,000 QPS	15,000 – 50,000 QPS

METRIC	1,000 CONCURRENT USERS (CCU)	5,000 CONCURRENT USERS (CCU)	10,000 CONCURRENT USERS (CCU)
Active Session Memory (Redis)	~250 MB in RAM	~1.2 GB in RAM	~2.5 GB in RAM
Network Throughput (CDN Offload)	150 – 350 Mbps	750 Mbps – 1.8 Gbps	1.5 Gbps – 3.8 Gbps

3. Complete Infrastructure Stack & Bill of Materials (BOM)

Everything needed from Load Balancers, Redis, MongoDB/PostgreSQL, Compute clusters to CDN and Security.

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+-----------------------------------------------------------------------------------------+ | GLOBAL CLIENT TRAFFIC |
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+──────────────────────────────────────────────+ | Cloudflare Enterprise / Pro Edge (WAF + CDN) | | - DDoS Mitigation & Rate
Limiting at Edge | | - Static & 3D Asset Caching (Zero-Egress R2/S3) | +──────────────────────────────────────────────+ | ▼
+──────────────────────────────────────────────+ | Layer 7 Application Load Balancer (ALB / NGINX) | | - SSL Termination +
HTTP/2 & HTTP/3 + Healthcheck| +──────────────────────────────────────────────+ |
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Next.js SSR #1 | | Next.js SSR #2 | | Next.js SSR #N | | (Auto-Scaled) | | (Auto-Scaled) | | (Auto-Scaled) | +────────┬────────+
+────────┬────────+ +────────┬────────+ | | | ├─────────────────────┼─────────────────────┤ | | ▼ ▼
+────────────────────────+ +────────────────────────+ | Redis Cluster (InMemory) | | PgBouncer Connection | | - User Sessions
& Tokens | | Pooler (10k+ Conns) | | - Sliding Rate Limits | +────────────┬────────────+ | - BullMQ Async Job Queue | |
+────────────────────────+ ▼ +────────────────────────────+ | Primary PostgreSQL / MongoDB | | (ACID Writes & Master
Store) | +────────────┬────────────+ | (Async Stream) ▼ +────────────────────────────+ | Read Replicas (1x to 3x
Nodes) | | (Lesson Reads, Cert Verification)| +────────────────────────────+
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Database Selection: PostgreSQL vs MongoDB for FinGenIQ

CRITERION	POSTGRESQL (RECOMMENDED FOR FINGENIQ)	MONGODB ATLAS (ALTERNATIVE)
Data Model Fit	**Ideal** Strict relational schema: Users, Sessions, Audit Logs, Progress, Certs, Renewals with foreign keys.	**Flexible** Good for unstructured lesson metadata, but requires application-level integrity for relations.
10k CCU Concurrency	**Excellent with PgBouncer** Handles 15k+ QPS easily with Read Replicas & MVCC.	**Excellent** High write throughput with Replica Sets & Sharded Clusters.
Audit & Compliance	**ACID Strict** Perfect for financial certifications, non-repudiation audit trails, and strict credentials.	**Document-level ACID** Requires multi-document transactions which increase latency under heavy load.
Monthly Cost at 10k CCU	**$650 – $1,200 / mo** (AWS RDS Aurora / Managed PG)	**$750 – $1,400 / mo** (MongoDB Atlas M40/M50 cluster)

Complete Component-by-Component Infrastructure Specification

INFRASTRUCTURE LAYER	COMPONENT & SOFTWARE	1,000 CCU SIZING	10,000 CCU SIZING	CORE FUNCTION
Edge & CDN	Cloudflare WAF / CDN + R2	Cloudflare Pro + 500GB R2	Cloudflare Enterprise / Business	DDoS shield, bot protection, 3D asset caching
Load Balancers	AWS ALB / HAProxy / NGINX Plus	1x Managed ALB	2x Redundant L7 ALBs	SSL termination, health checks, traffic routing
App Compute (Next.js)	Docker Containers on ECS / K8s	4x Nodes (2 vCPU, 4GB RAM)	16x – 24x Nodes (4 vCPU, 8GB RAM)	Next.js SSR, React 19, Server Actions, API routes
In-Memory Caching	Redis 7.2 Cluster / Sentinel	Primary + Replica (4GB RAM)	3-Node Sharded Cluster (16GB RAM)	Opaque session validation, IP rate limits, locks
Connection Pooler	PgBouncer (Transaction mode)	1x Container (Pool = 1,500)	2x Dedicated Nodes (Pool = 15,000+)	Prevents PostgreSQL connection exhaustion
Primary Database	PostgreSQL 16 / MongoDB Atlas	4 vCPU, 16GB RAM, 3k IOPS	16 vCPU, 64GB RAM, 10k+ IOPS	User accounts, auth logs, progress writes
Database Replicas	Streaming Read Replicas	1x Read Replica (4 vCPU, 16GB)	2x – 3x Read Replicas (8 vCPU, 32GB)	Offloads lesson reads, cert queries, admin searches

INFRASTRUCTURE LAYER	COMPONENT & SOFTWARE	1,000 CCU SIZING	10,000 CCU SIZING	CORE FUNCTION
Async Task Workers	BullMQ + Node.js Workers	2x Worker Instances	6x Worker Instances	PDF cert generation, bulk emails, webhook dispatches
Transactional Email	AWS SES / Resend / Postmark	~50,000 emails/mo	~500,000 emails/mo	Activation tokens, password resets, renewal reminders
APM & Observability	Datadog / Sentry / Prometheus	Basic metrics & Sentry APM	Full APM, distributed tracing, log indexing	Real-time alerts, error detection, latency monitoring

4. Monthly Operating Expense Audit (OpEx)

INFRASTRUCTURE COMPONENT	1,000 CCU (AWS MANAGED)	1,000 CCU (HETZNER VPS)	10,000 CCU (AWS MANAGED)	10,000 CCU (HETZNER BARE-METAL)
Next.js App Compute	$160 / mo	$65 / mo	$1,850 / mo	$520 / mo
Database (Primary + Replicas)	$280 / mo	$110 / mo	$2,200 / mo	$680 / mo
Redis Cache & BullMQ	$65 / mo	$25 / mo	$380 / mo	$120 / mo
Load Balancers (ALB / Ingress)	$45 / mo	$15 / mo	$240 / mo	$50 / mo
Cloudflare CDN, WAF & R2 Storage	$35 / mo	$35 / mo	$450 / mo	$250 / mo
Transactional Email Delivery	$40 / mo	$40 / mo	$350 / mo	$350 / mo
APM, Metrics & Sentry Logs	$90 / mo	$40 / mo	$650 / mo	$200 / mo
Cold Backups & Disaster Recovery	$15 / mo	$10 / mo	$80 / mo	$30 / mo
TOTAL MONTHLY OPEX	**$730 / mo**	**$340 / mo**	**$6,200 / mo**	**$2,200 / mo**
ANNUALIZED INFRASTRUCTURE	**$8,760 / yr**	**$4,080 / yr**	**$74,400 / yr**	**$26,400 / yr**

Unit Economics Summary

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At 1,000 CCU (~75,000 Registered Users): Cost = $730 / 75,000 = $0.0097 per user / month ($0.11 / user / year)
At 10,000 CCU (~750,000 Registered Users): Cost = $6,200 / 750,000 = $0.0082 per user / month ($0.098 / user / year)
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5. Critical Codebase Refactors Required for 10k CCU

1. SQLite to PostgreSQL / MongoDB Migration (Mandatory)
The current `better-sqlite3` implementation will hard-lock with `SQLITE_BUSY` errors under simultaneous write transactions. Replace with PostgreSQL using connection pooling (PgBouncer) and indexed foreign keys.

2. In-Memory Session & Rate Limiting Offload
Currently, FinGenIQ queries `sessions` and writes to `ip_rate_limits` on disk for every HTTP request. Moving these to a **Redis Cluster** reduces latency from ~25ms disk I/O to < 0.8ms in-memory lookups.

3. Three.js Static Asset CDN Offloading
Ensure all WebGL geometries, textures, and heavy client JavaScript chunks have strict caching headers (`max-age=31536000, immutable`) so that 10,000 simultaneous users download 3D assets directly from Cloudflare Edge rather than hitting your Next.js application servers.

4. Asynchronous Queue for Certificate & Email Generation

> Cryptographic SHA-256 certificate generation and transactional emails should be offloaded to **BullMQ workers** rather than processed synchronously during learner completion requests.